PRETHIVIKA G (192521085)

ROOBINI P (192525055)

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COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	B .Tech (IT, AIML), B.E(CSE WITH AI)
Programme:	Bachelor in Technology /Engineering
Course Code & Course Name	CSA1503-Cloud Computing and big Data Analysis
Academic Year / Batch	2025
Faculty Name	Dr.A.Gnana Soundari
Assignment Title	Design and Critical Analysis of a Cloud-Based Service Architecture for a Scalable Online Education Organization
Date of Issue	02-09-2026
Date of Submission	02-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO1: Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions. CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. CO3: Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.
Bloom's Taxonomy Level	L3 – Apply; L4 – Analyse
SDG Mapping (SDG 1-17)	SDG 4 – Quality Education SDG 9 – Industry, Innovation and Infrastructure SDG 12 – Responsible Consumption and Production
Industry / Societal Relevance	Enables scalable, cost-effective, and data-driven online education through cloud computing, virtualization, web services, and big data analytics.

B. Assignment Problem / Challenge

- Design and critically analyze a **cloud-based service architecture for a scalable online education organization** that delivers online courses, live classes, recorded lectures, assessments, assignments, and learning resources to a large number of students.
- The proposed system should support **thousands of concurrent users**, provide high availability, ensure secure storage of student and academic data, and automatically scale resources according to demand. The architecture should use appropriate **cloud computing services, distributed storage, databases, content delivery, load balancing, monitoring, and security mechanisms**.
- The system should efficiently handle peak workloads such as **online examinations, assignment submissions, live classes, and course registrations** while maintaining good performance and minimizing operational cost.
- During peak periods such as **course registration, live classes, online examinations, and assignment submission deadlines**, the number of users increases significantly. The existing system experiences problems such as **server overload, slow response time, insufficient storage, service downtime, and difficulty in managing large volumes of educational data**.
- The organization wants to migrate its services to a **cloud-based architecture** that can dynamically scale according to the number of users and workload. The proposed architecture must provide **high availability, scalability, security, fault tolerance, efficient data storage, low latency, and cost effectiveness**.
- The system should support students, faculty members, administrators, and content managers while providing secure access to educational resources

C. Problem Statement

Problem / Case / Design Challenge:

- A growing online education organization is planning to modernize its IT infrastructure to support an increasing number of students, teachers, online courses, examinations, video lectures, and digital learning services.
- The organization currently maintains physical servers and locally installed software.
- As the number of user’s increases, it faces problems such as
- high infrastructure cost, limited scalability, hardware maintenance, software
- deployment difficulties, and inefficient utilization of computing resources.
- The organization plans to adopt cloud computing and cloud-based services to improve flexibility, scalability, resource utilization, and service availability. The
- proposed solution should consider cloud computing, its evolution, hardware and
- Internet software evolution, server virtualization, web services, and cloud service models.

D. Requirements and Constraints

Requirements

Requirement	Description
Scalability	Support thousands of students simultaneously.
Availability	Provide continuous and reliable service.
Performance	Deliver courses and videos with low latency.
Security	Protect student and academic data.
Storage	Store videos, assignments and student records securely.
Backup	Provide regular data backup and recovery.

Constraints

- Must handle peak traffic during exams and live classes.
- Requires large storage for videos and learning materials.
- Depends on reliable internet connectivity.
- Must maintain low latency for online learning.
- Cloud resources should remain cost-efficient.

E. Student Work

1. Problem Understanding and Formulation

Problem:
Traditional online education platforms may face scalability, availability, performance, and reliability issues when a large number of students access courses, videos, assignments, and examinations simultaneously.

Expected Outcomes:

- Scalable cloud-based education architecture
- High availability and reliability
- Support for large numbers of concurrent users
- Secure student and course data
- Efficient content delivery
- Cost-effective resource utilization

Available Information/Data:

- Student and instructor information
- Course and enrollment data
- Video and learning content
- Assignment and examination data
- User activity and performance data
- Cloud computing resources

Assumptions:

- Students have internet connectivity.
- Cloud infrastructure is available.
- Users authenticate before accessing protected resources.
- Learning content is stored in cloud storage.

Constraints:

- Budget and cloud-resource limitations
- Network bandwidth
- Security and privacy requirements
- Response-time requirements
- Scalability requirements
- Data availability and reliability

2. Application of Course Knowledge

Relevant cloud-computing concepts include:

- Cloud service models — **IaaS, PaaS, SaaS**
- Virtualization
- Elasticity
- Scalability
- Load balancing
- Auto-scaling
- Distributed computing
- Cloud storage
- Database services
- Content Delivery Network (CDN)
- Fault tolerance
- High availability
- Cloud security
- Monitoring

Important performance metrics:

Availability:

$$Availability = \frac{Uptime}{Uptime + Downtime} \times 100$$

Throughput:

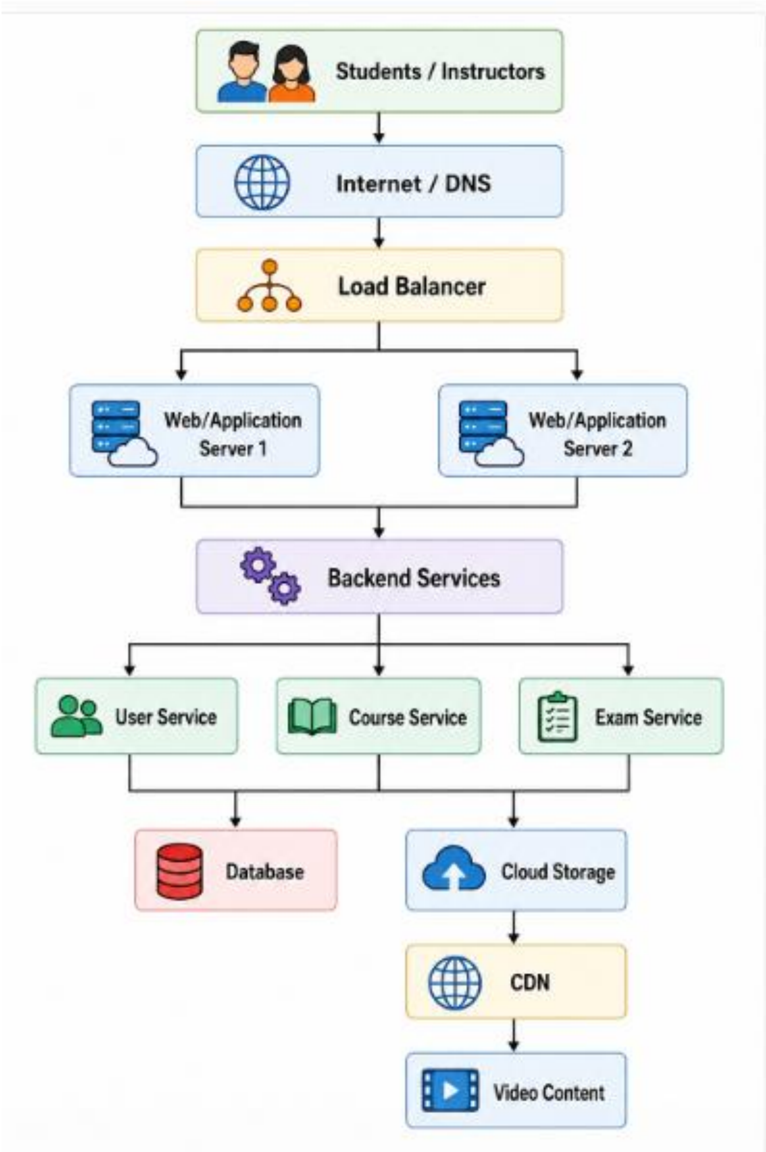
$$Throughput = \frac{Number\ of\ Requests}{Time}$$

Response Time:

$$Response\ Time = Processing\ Time + Network\ Time$$

3. Solution / Design / Methodology

The proposed architecture can follow this structure:



4. Use of Modern Tools

For this project, you can use:

- **AWS / Microsoft Azure / Google Cloud**
- Docker
- Kubernetes
- Python
- MySQL/PostgreSQL
- Cloud storage
- Load balancer
- Auto-scaling
- CDN
- Git/GitHub
- Draw.io for architecture diagrams
- Cloud monitoring tools

For a college-level project, **AWS architecture + Draw.io + Python/Docker** would be a strong combination.

5. Results and Validation

You should present results such as:

Metric	Without Scaling	With Cloud Scaling
Response Time	Higher	Lower
Concurrent Users	Limited	High
Availability	Moderate	High
Resource Utilization	Fixed	Dynamic
Fault Tolerance	Low	High
Cost Efficiency	Lower	Better

You can also include:

- Architecture screenshots
- Cloud deployment screenshots
- Load-testing results
- Response-time graph
- Number of concurrent users
- CPU/memory utilization
- Availability results
- Scaling demonstration

6. Analysis and Engineering Decision

Compare different architectures:

Approach	Scalability	Cost	Complexity	Reliability
Traditional Server	Low	Medium	Low	Low
VM-Based Cloud	Medium	Medium	Medium	Medium
Cloud + Auto Scaling	High	Optimized	Medium	High
Serverless/Microservices	Very High	Pay-per-use	High	High

7. Broader Considerations

Discuss:

- **Sustainability:** Dynamic resource allocation reduces unnecessary infrastructure usage.
- **Environment:** Cloud resource sharing can reduce physical infrastructure requirements.
- **Society:** Enables students to access education remotely.
- **Safety:** Authentication, encryption, backups, and access control protect users.
- **Ethics:** Student data must be handled responsibly.
- **Economics:** Pay-as-you-go cloud services can reduce initial infrastructure costs.
- **Accessibility:** The platform should support users with different devices and network conditions.
- **Professional responsibility:** The architecture must be designed for security, reliability, and privacy.

8. Conclusion

The proposed cloud-based service architecture provides a **scalable, reliable, secure, and cost-effective platform for online education**. The use of load balancing, auto-scaling, cloud storage, CDN, managed databases, and monitoring allows the organization to handle varying numbers of users and learning workloads. The architecture can improve availability and performance while reducing the limitations of traditional fixed infrastructure.

9. Student Reflection

ROOBINI P (192525055)

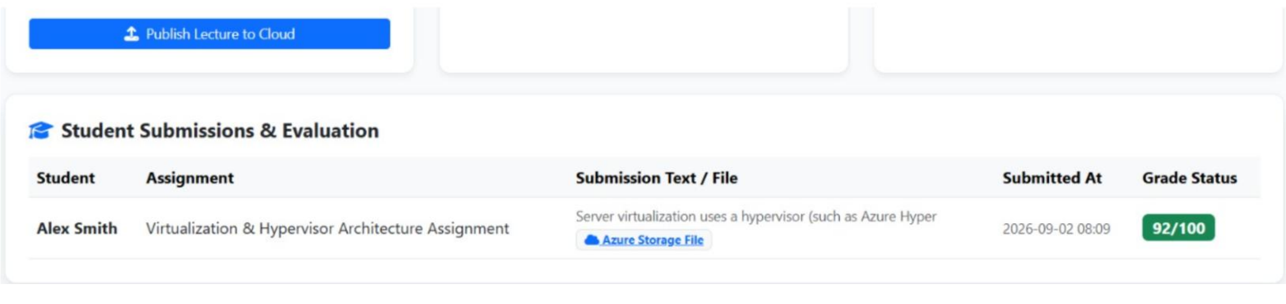
Working on this project helped me understand how cloud computing concepts can be applied to solve real-world problems in the education sector. I learned about IaaS, PaaS, SaaS, XaaS, server virtualization, and web services and understood how these technologies work together to build a scalable online education platform. Developing the web application also improved my practical skills in designing system architecture, connecting the frontend with backend services, and managing data. This project helped me connect the theoretical concepts learned in class with their practical implementation.

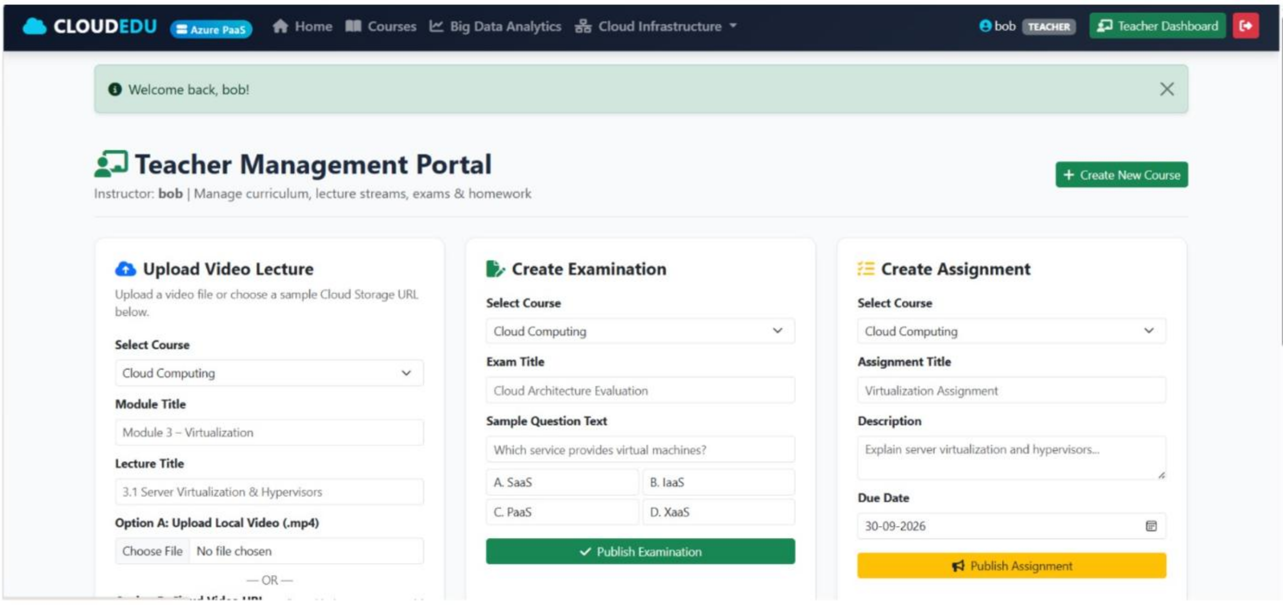
PRETHIVIKA G (192521085)

This project gave me practical experience in designing a cloud-based system for an organization with continuously increasing users and data. I learned how virtualization can improve resource utilization and how cloud services can reduce infrastructure and maintenance costs. Implementing features such as course management, online examinations, cloud storage, and analytics helped me understand the importance of scalability and availability in modern applications. I also gained confidence in analyzing different cloud service models and selecting suitable services according to organizational requirements.

OAVIKSHA P L (192572001)

Through this project, I developed a better understanding of how cloud computing and big data analytics can work together in an online education environment. I learned how student-related data such as course activity, examination results, and learning progress can be analyzed to generate useful insights. Designing the analytics dashboard improved my knowledge of data visualization and data-driven decision-making. Overall, the project strengthened my technical, problem-solving, and system-design skills and gave me a better understanding of how modern cloud technologies are used in real-world educational platforms.

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3.1 Server Virtualization & Hypervisors
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Exam Title
Cloud Architecture Evaluation
Sample Question Text
Which service provides virtual machines?
A. SaaS
B. IaaS
C. PaaS
D. XaaS
Publish Examination
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Description
Explain server virtualization and hypervisors...
Due Date
30-09-2026
Publish Assignment

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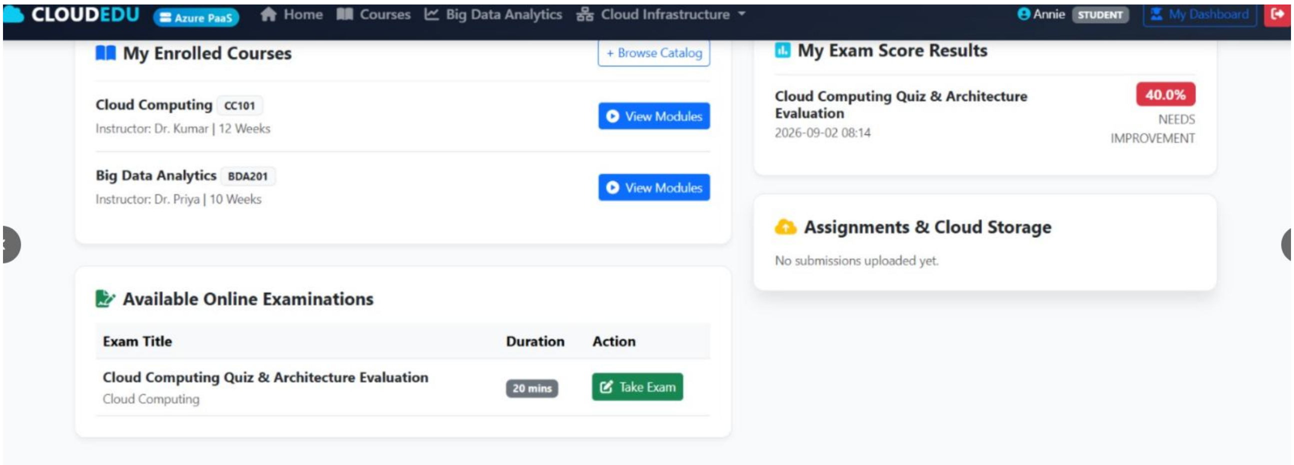
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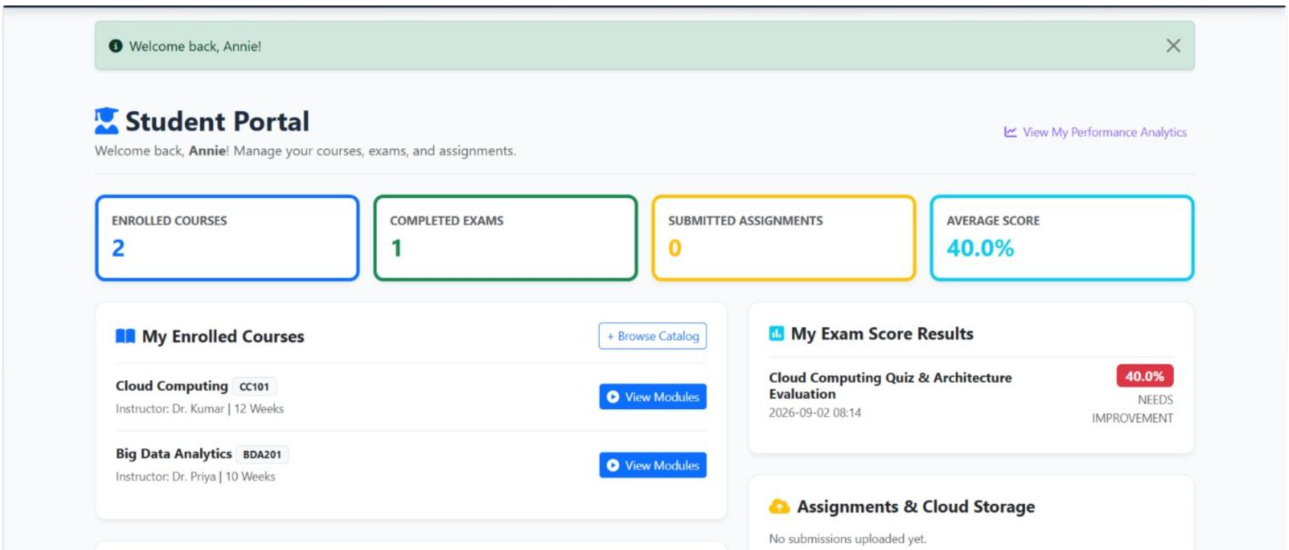
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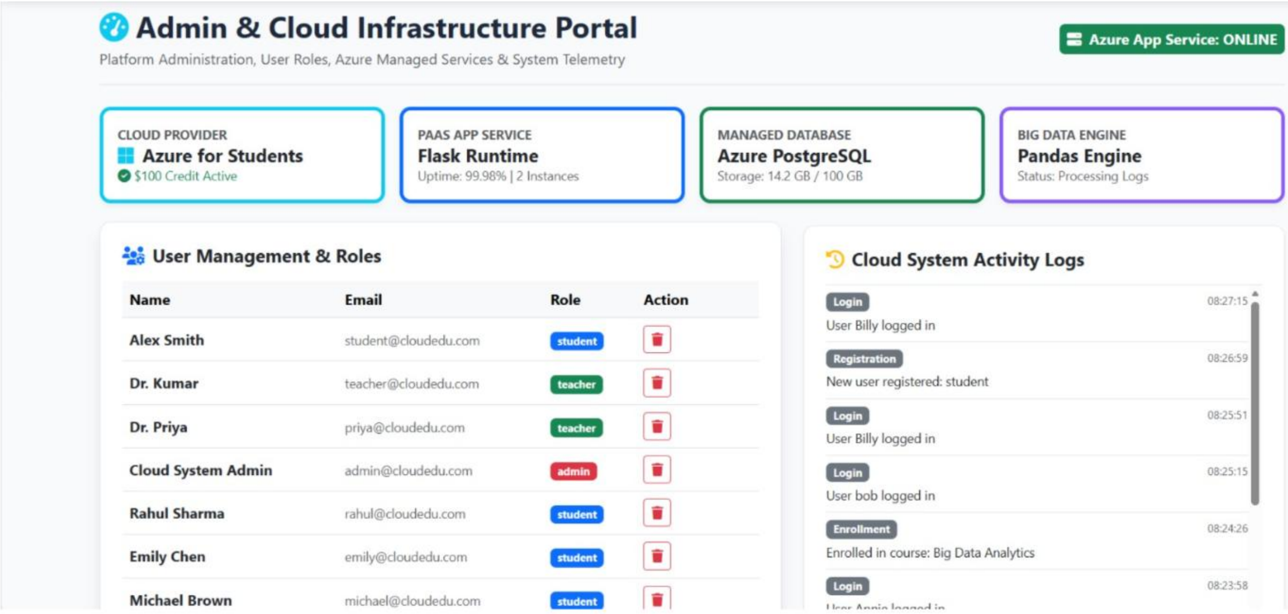
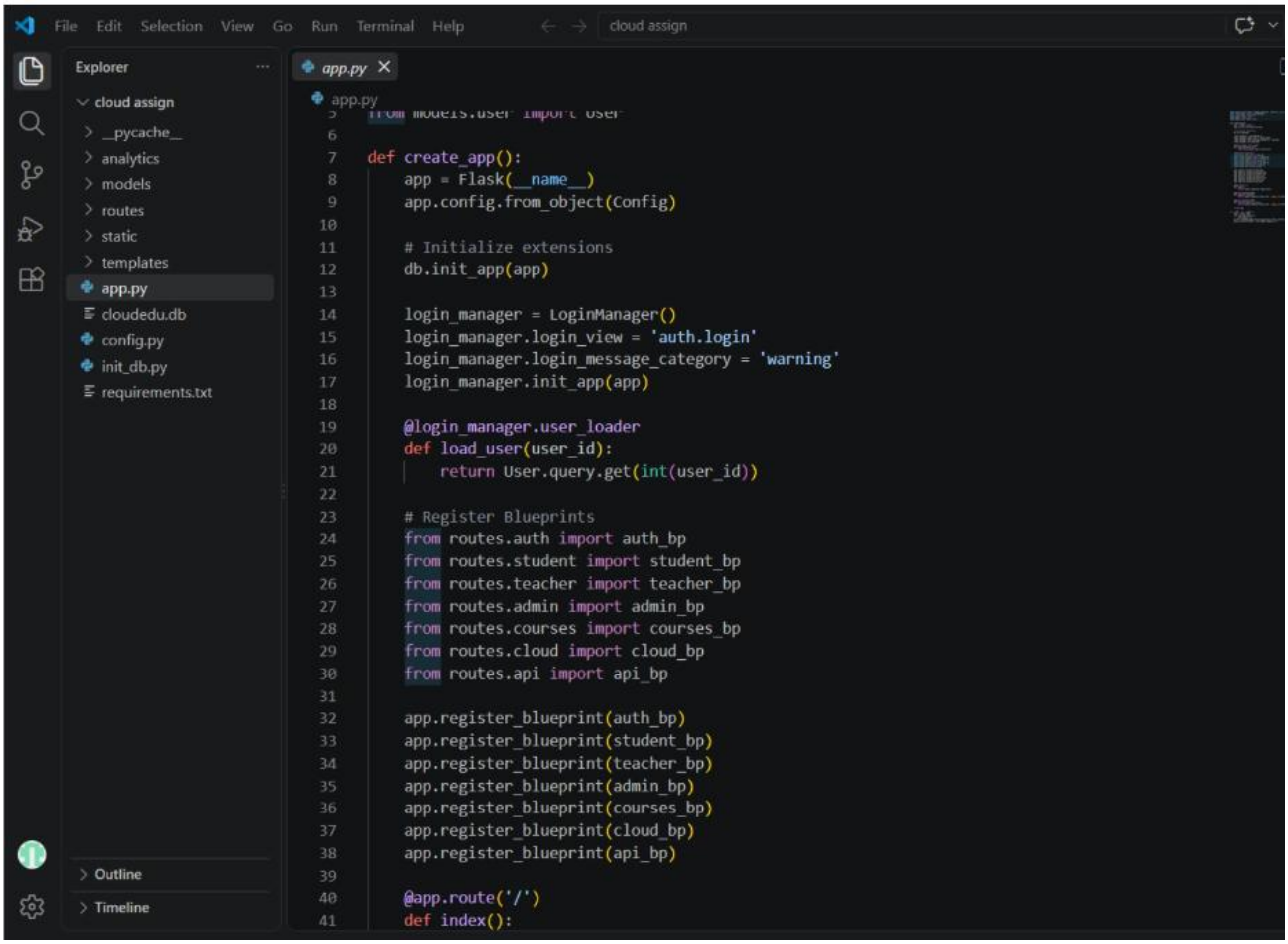


SOURCE CODE:




```python
from models.user import User

def create_app():
    app = Flask(__name__)
    app.config.from_object(Config)

    # Initialize extensions
    db.init_app(app)

    login_manager = LoginManager()
    login_manager.login_view = 'auth.login'
    login_manager.login_message_category = 'warning'
    login_manager.init_app(app)

    @login_manager.user_loader
    def load_user(user_id):
        return User.query.get(int(user_id))

    # Register Blueprints
    from routes.auth import auth_bp
    from routes.student import student_bp
    from routes.teacher import teacher_bp
    from routes.admin import admin_bp
    from routes.courses import courses_bp
    from routes.cloud import cloud_bp
    from routes.api import api_bp

    app.register_blueprint(auth_bp)
    app.register_blueprint(student_bp)
    app.register_blueprint(teacher_bp)
    app.register_blueprint(admin_bp)
    app.register_blueprint(courses_bp)
    app.register_blueprint(cloud_bp)
    app.register_blueprint(api_bp)

    @app.route('/')
    def index():
```

10. References

Use **IEEE referencing style** for:

- M. Armbrust et al., “A View of Cloud Computing,” Communications of the ACM, vol. 53, no. 4, pp. 50–58, 2010.
- P. Mell and T. Grance, The NIST Definition of Cloud Computing, NIST Special Publication 800-145, National Institute of Standards and Technology, 2011.
- R. Buyya, C. S. Yeo, S. Venugopal, J. Broberg, and I. Brandic, “Cloud Computing and Emerging IT Platforms: Vision, Hype, and Reality for Delivering Computing as the 5th Utility,” Future Generation Computer Systems, vol. 25, no. 6, pp. 599–616, 2009.
- R. Buyya, J. Broberg, and A. V. Dastjerdi, Cloud Computing: Principles and Paradigms. Hoboken, NJ, USA: Wiley, 2011.
- T. Erl, R. Puttini, and Z. Mahmood, Cloud Computing: Concepts, Technology & Architecture. Upper Saddle River, NJ, USA: Prentice Hall, 2013.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	**100**